

EXERCISES

Let $(\Omega, \mathcal{F}, P)$ be a complete probability space.

1. Let X and Y be two integrable r.v., $\mathcal{G}$ be a sub σ -field of $\mathcal{F}$. Suppose that $\sigma(X)$, $\sigma(Y)$ and $\mathcal{G}$ are independent. Show that

$$E(XY|\mathcal{G}) = E(X)E(Y), \text{ a.s.}$$

2. Let $\underline{X} = (X_1, \dots, X_d)$ be a vector of real r.v.

- (a) Suppose $X_1, \dots, X_d$ to be independent Gaussian r.v. Show that the vector $\underline{X} = (X_1, \dots, X_d)$ is Gaussian.
- (b) Suppose that $\underline{X}$ is a Gaussian vector. Show that $X_1, \dots, X_d$ are independent if and only if the matrix $\Gamma(\underline{X})$ is diagonal.
- (c) Suppose that $\underline{X}$ be a square integrable random vector. Prove that the covariance matrix Γ is a positive semidefinite symmetric matrix.

From now on let $(\mathcal{F}_t)$ be a filtration fulfilling the usual conditions. W will denote an $(\mathcal{F}_t)$ -standard Brownian motion. Let $(M_t; 0 \leq t \leq T)$ be an $(\mathcal{F}_t)$ -martingale such that $E[M_T^2] < \infty$. Prove the following.

3. (a) For any $t \in [0, T]$, $E[M_t^2] < \infty$ and the function $t \mapsto E[M_t^2]$ is non-decreasing.
- (b) For any s, t in $[0, T]$,

$$E[(M_t - M_s)^2 | \mathcal{F}_s] = E[M_t^2 | \mathcal{F}_s] - M_s^2, \quad \text{if } 0 \leq s \leq t \leq T.$$

4. Let $X = (X_t)_{t \in [0, T]}$ be a càdlàg $(\mathcal{F}_t)$ -local martingale and τ be a stopping time. Show that X^τ is still a local martingale.
5. Let $X = (X_t)_{t \in [0, T]}$ be a $(\mathcal{F}_t)$ -supermartingale such that $E(X_T) = E(X_0)$. Show that X is an $(\mathcal{F}_t)$ -martingale.
6. Let $X = (X_t)_{t \in [0, T]}$ be an $(\mathcal{F}_t)$ -local martingale such that $X_0 \in \mathcal{L}^1$. Show the following. If X is non-negative, then it is a supermartingale.
7. **(Pricing of a European call in Black-Scholes model.)**
Particular case: discount rate $r = 0$.

Let $\sigma > 0$, $s_0 > 0$. We set

$$S_t = s_0 \exp \left(\sigma W_t - \frac{\sigma^2}{2} t \right).$$

$(\mathcal{F}_t)$ is here the canonical filtration associated with W . Let $T > 0$, $r > 0$. We fix $t \in [0, T]$.

i) Evaluate

$$V_t = E(f(S_T) | \mathcal{F}_t)$$

in the form

$$V_t = F(t, S_t).$$

ii) Verify that

$$F(t, x) = \int f(x e^{-\frac{\sigma^2}{2}\theta + \sigma y \sqrt{\theta}}) p(y) dy,$$

where $\theta = T - t$ and p is the density of $N(0, 1)$.

iii) We suppose $f(x) = (x - K)_+$ for some $K > 0$. We set

$$d_1 = \frac{\log(\frac{K}{x}) + \frac{\sigma^2 \theta}{2}}{\sigma \sqrt{\theta}}.$$

Express $F(t, x)$ using the distribution function ψ of $N(0, 1)$ law.

8. Let $B = (W^1, \dots, W^d)$ be a d -dimensional standard Brownian motion and $(\mathcal{F}_t^B)$ be its canonical filtration. In particular $W^1, \dots, W^d$ are independent standard Brownian motions.

(a) Prove that $\text{Cov}(W_t^i, W_t^j) = \delta_{ij}t$, $1 \leq i, j \leq d$.

(b) Suppose $i \neq j$. Prove that, for $s < t$, we have

$$E(W_t^i W_t^j | \mathcal{F}_s) = W_s^i W_s^j, \text{ a.s.}$$

(c) Show that for every $u = (u^1, \dots, u^d) \in \mathbb{R}^d$.

$$E \left(e^{u \cdot B_t - \frac{1}{2} \|u\|^2 t} | \mathcal{F}_s \right) = e^{u \cdot B_s - \frac{1}{2} \|u\|^2 s}.$$

9. Let $\alpha > -\frac{1}{2}$ and let $(I_t)_{t \geq 0}$ be the process defined by

$$I_t = \int_0^t |W_s|^\alpha dW_s.$$

Show that it is an $(\mathcal{F}_t)$ -martingale.

10. Let X and $\bar{X}$ be two Itô processes of the type

$$\begin{aligned} X_t &= x + \int_0^t \sigma_s dW_s + \int_0^t b_s ds \\ \bar{X}_t &= \bar{x} + \int_0^t \bar{\sigma}_s dW_s + \int_0^t \bar{b}_s ds, \end{aligned}$$

where $x, \bar{x} \in \mathbb{R}$, $\sigma, b, \bar{\sigma}, \bar{b}$ are $(\mathcal{F}_t)$ -adapted measurable processes. Show that $X\bar{X}$ is a Itô process and give its canonical decomposition.

11. Let M be a continuous $(\mathcal{F}_t)$ -square integrable martingale. Let A be the increasing adapted process vanishing at zero intervening in the Doob-Meyer decomposition of the submartingale M^2 . Let $0 \leq a_0 \leq a \leq b \leq u \leq v$. We have the following.

(a) $E((M_v - M_u)(M_b - M_a)|\mathcal{F}_{a_0}) = 0$.

(b)

$$E[(M_b - M_a)^2|\mathcal{F}_{a_0}] = E[M_b^2 - M_a^2|\mathcal{F}_{a_0}] = E(A_b - A_a|\mathcal{F}_{a_0});$$

12. Let X be a continuous $(\mathcal{F}_t)$ -semimartingales, Z be an $(\mathcal{F}_t)$ -adapted measurable cadlag process.

Show that the Itô integrals $\int_0^t Z dX$ and $\int_0^t Z^- dX$ are equal, where $Z_s^- = Z_{s-}$, $s \geq 0$.

13. a) Evaluate the quadratic variation of $[\sin(W), \sin(W)]$. Let A be a zero quadratic variation process and set $D = W + A$. Deduce $[\sin(D), \sin(D)]$.

b) If X is a Itô process of the form

$$X = X_0 + \int_0^\cdot H_s dW_s + \int_0^\cdot K_s ds,$$

express $\int_0^t X \circ dW$ in terms of Itô integrals.

c) Suppose that F is of class $C^2(\mathbb{R})$. Verify that

$$M_t = F(W_t) - \frac{1}{2} \int_0^t F''(W_s) ds,$$

is an $(\mathcal{F}_t)$ -martingale if $\int_{\mathbb{R}} |F'|^2(y) \frac{p(\frac{y}{\sqrt{t}})}{1+y^2} dy < \infty$, where p is the law density of $N(0, 1)$.

14. Let (X_t) be an $(\mathcal{F}_t)$ -adapted measurable process and $T > 0$ such that $E(\int_0^T |X_s|^{2p} ds) < \infty$ for some $p \geq 1$.

(a) Applying Itô's formula to $M_t = \int_0^t X_s dW_s$, show the existence of an increasing sequence (T_n) of stopping times diverging to ∞ such that

$$E(|M_{T \wedge T_n}^{2p}|) = p(2p-1)E\left(\int_0^{T \wedge T_n} |M_t|^{2(p-1)} X_t^2 dt\right).$$

(b) Is the function $t \mapsto E(|M_{t \wedge T_n}|^{2p})$ monotone? Deduce that

$$E(|M_{T \wedge T_n}^{2p}|) \leq [p(2p-1)]^p T^{p-1} E\left(\int_0^{T \wedge T_n} |X_s|^{2p} ds\right)$$

and then

$$E(|M_T^{2p}|) \leq [p(2p-1)]^p T^{p-1} E\left(\int_0^T |X_s|^{2p} ds\right).$$

(c) Applying Doob's inequality show the existence of a constant K_p that we will express explicitly such that

$$E\left(\sup_{0 \leq s \leq T} |M_t|^{2p}\right) \leq K_p E\left(\int_0^T |X_s|^{2p} ds\right).$$

15. **Ornstein-Uhlenbeck process.** Let $x_0, \sigma, b \in \mathbb{R}$. Let Z be a continuous process vanishing at zero.

(a) Show that whenever X^1, X^2 are solutions of

$$X_t = x_0 + Z_t - b \int_0^t X_s ds, t \in [0, T]. \quad (1)$$

Then $X^1 = X^2$.

(b) Show that the process $I := \int_0^\cdot e^{bs} d^- Z_s$ is well-defined and it also equals $\int_0^\cdot e^{bs} d^\circ Z_s$.

(c) We set

$$X_t := e^{-bt} \left(x_0 + \int_0^t e^{bs} d^- Z_s \right). \quad (2)$$

Show that X solves equation (1).

(d) Deduce that X is the unique solution of (1).

(e) Suppose that Z is a mean-zero square integrable process with independent increments and such that $\text{Var}(Z_t - Z_s) = \psi(t) - \psi(s), t \geq s$, where $\psi : \mathbb{R}_+ \rightarrow \mathbb{R}$ is a continuous function vanishing at zero.

i. Show that ψ is non-decreasing.

ii. Show that Z is a martingale and calculate its quadratic variation.

iii. Show that $E(I_t) = 0$ and $\text{Var}(I_t) = \int_0^t e^{2bs} d\psi(s)$, for every $t \geq 0$.

iv. Calculate $E(X_t)$ and $\text{Var}(X_t), t \geq 0$.

(f) Suppose Z to be equal to σW . Determine X in terms of Itô integral.

(g) Show that the process X is Gaussian.

(h) Determine the law of X_t for any $t \geq 0$.

16. Let $\alpha \in \mathbb{R}$ be a constant. Consider the stochastic differential equation

$$\begin{cases} dX_t &= -\alpha^2 X_t^2(1 - X_t)dt + \alpha X_t(1 - X_t)dW_t \\ X_0 &= x_0 \in]0, 1[. \end{cases} \quad (3)$$

- (a) Do the coefficients of (3) fulfill the usual assumptions of the existence and uniqueness theorem?
- (b) We admit that a solution X takes its values in the interval $]0, 1[$. We set $Y_t = \frac{X_t}{1-X_t}$. Determine the stochastic differential equation fulfilled by Y .
- (c) Deduce that such a solution of (3) gives

$$X_t = \frac{x_0 \exp(\alpha W_t - \frac{\alpha^2 t}{2})}{x_0 \exp(\alpha W_t - \frac{\alpha^2 t}{2}) + 1 - x_0}.$$

17. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function.

- (a) Suppose that f is bounded. Let $u : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$u(t, x) = \begin{cases} \int_{\mathbb{R}} p_{T-t}(x - y) f(y) dy, & : t \in [0, T[\\ f(x) & : t = T, \end{cases} \quad (4)$$

where p_s is the law density of $N(0, s)$, $s > 0$. Show that u is a continuous function, it is of class $C^{1,2}([0, T[\times \mathbb{R})$ and $v = u$ solves

$$\partial_t v + \frac{1}{2} \partial_{xx}^2 v = 0, v(T, x) = f(x). \quad (5)$$

- (b) Show that for every $p \geq 1$, $E(\sup_{t \leq T} |W_t|^p) < \infty$.
- (c) Let $v : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function of class $C^{1,2}([0, T[\times \mathbb{R})$. We suppose moreover that v has at most polynomial growth in the sense that

$$\sup_{t \leq T, x \in \mathbb{R}} |v(t, x)| \leq \text{const}(1 + |x|^p), \quad (6)$$

for some $p \geq 1$.

Show that $h_0 = v(0, 0)$, $Z_t = \partial_x v(t, W_t)$ provides the predictable decomposition of the r.v. $h = f(W_T)$, i.e. $h = h_0 + \int_0^T Z_s dW_s$.

Hint. Apply first Itô formula to $v(t, W_t)$, $0 \leq t < T$. Then use the predictable representation theorem for Brownian martingales.

18. Let α, β two positive constants.

(a) Let $\lambda \in \mathbb{R}$. Show that the process

$$L_t := \exp \left(-\lambda \frac{W_t^2 - t}{2} - \frac{\lambda^2}{2} \int_0^t W_s^2 ds \right),$$

is an $(\mathcal{F}_t)$ -martingale.

Hint. Apply the proposition due to Benes with $b(s, x) = -\lambda x$ and Itô formula applied to W_t^2 .

(b) Let Q be the probability on $(\Omega, \mathcal{F}_T)$ defined by $dQ = L_T dP$. Deduce that under Q $W_t = B_t - \lambda \int_0^t W_s ds$ is an Ornstein-Uhlenbeck process. Show that under Q W_t is distributed according to $N(0, \frac{1-e^{-2\lambda t}}{2\lambda})$.

(c) We set

$$I := E \left(\exp \left(-\alpha W_T^2 - \frac{\lambda^2}{2} \int_0^T W_s^2 ds \right) \right). \quad (7)$$

Deduce that

$$I = E^Q \left(\exp(-\alpha W_T^2 + \frac{\lambda}{2}(W_T^2 - T)) \right) = E \left(\exp(-\alpha X_T^2 + \frac{\lambda}{2}(X_T^2 - T)) \right),$$

where X is an Ornstein-Uhlenbeck solving

$$X_t = W_t - \lambda \int_0^t X_s ds, t \geq 0.$$

(d) Show that $I = (\cosh(\lambda T) + \frac{2\alpha}{\lambda} \sinh(\lambda T))^{-\frac{1}{2}}$.

(e) Conclude that, for every $\beta \geq 0$

$$E \left(\exp \left(-\beta \int_0^T W_s^2 ds \right) \right) = (\cosh(\sqrt{2\beta} T))^{-\frac{1}{2}}.$$

19. **Martingale problem.** Let $a : [0, T] \times \mathbb{R}^m \rightarrow \mathbb{R}^{m \times p}$, $b : [0, T] \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be Borel functions. We denote by $E(a, b)$ the stochastic differential equation

$$dX_t = a(t, X_t)dW_t + b(t, X_t)dt. \quad (8)$$

- (a) Let X be a solution of $E(a, b)$. Show that for every $f \in C^2(\mathbb{R}^m)$ then

$$f(X_t) - f(X_0) - \int_0^t \nabla f(X_s) \cdot b(s, X_s) ds - \frac{1}{2} \sum_{i,j=1}^m \int_0^t \partial_{x_i, x_j}^2 f(s, X_s) (aa^*)_{ij}(s, X_s) ds \quad (9)$$

is a local martingale, where a^* is the transposed of matrix a .

We say in this case that (X, P) solves the **martingale problem** related to b and aa^* .

- (b) Suppose now (X, Q) solves previous martingale problem, where X a continuous process and Q is a probability on $(\Omega, \mathcal{F})$. Q is the reference probability for the items below.

- i. Show that the processes

$$M_t^i := X_t^i - \int_0^t b^i(s, X_s) ds, t \geq 0, 1 \leq i \leq m,$$

are $(\mathcal{F}_t^X)$ -local martingales under Q and $(\mathcal{F}_t^X)$ is the canonical filtration associated with X .

- ii. Show that for every $1 \leq i, j \leq m$, $[M^i, M^j] = [X^i, X^j]$.

- iii. Express (9) for $f(x) = x_i x_j$ to show that

$$[M^i, M^j]_t = \int_0^t (aa^*)_{ij}(s, X_s) ds.$$

- iv. Consider now for simplicity the case $m = p = 1$ and $a > 0$. Let $M = M^1$. Show that the process

$$W_t := \int_0^t \frac{1}{a(s, X_s)} dM_s, t \geq 0,$$

is an $(\mathcal{F}_t^X)$ -Brownian motion.

- v. Deduce finally that X is a solution of $E(a, b)$ and $E(a, b)$ admits weak existence.

20. Let $a : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$, $b : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be Borel functions with linear growth with respective constants c_a, c_b .

We denote again by $E(a, b)$ the stochastic differential equation (8) with $m = p = 1$. Let $\gamma \geq 2$.

Let $X = (X_t)_{t \in [0, T]}$ be a solution of $E(a, b)$, where $X_t \in L^\gamma(\Omega, \mathcal{F}, P)$ for all $t \in [0, T]$. Without restriction of generality we take $T = 1$. Show the existence of constants $C_i := C_i(\gamma, c_a, c_b, E(|X_0|^\gamma))$, $i = 1, 2$ with

- (a)

$$\sup_{t \leq T} E(|X_t|^\gamma) \leq C_1. \quad (10)$$

(b)

$$E(|X_t - X_s|^\gamma) \leq C_2 \cdot |t - s|^{\frac{\gamma}{2}}, \quad s, t \in [0, T]. \quad (11)$$

21. **Non-explosion.** Suppose $m = p = 1$. Let $Lf = \frac{1}{2}a^2(x) + b(x)$ where $a, b : \mathbb{R} \rightarrow \mathbb{R}$ are Borel function such that $a > 0$ and $a, b, \frac{b}{a^2}$ are locally integrable. Let $v : \mathbb{R} \rightarrow \mathbb{R}$ be the unique solution of

$$\begin{cases} Lv &= 1 \\ v(0) &= 0, \quad v'(0) = 0. \end{cases} \quad (12)$$

We denote $\Sigma(x) := 2 \int_0^x \frac{b}{a^2}(y) dy, x \in \mathbb{R}$.

- i) Verify that whenever

$$\int_0^\infty e^{-\Sigma(y)} dy = \int_{-\infty}^0 e^{-\Sigma(y)} dy = \infty, \quad (13)$$

then

$$v(-\infty) = v(+\infty) = +\infty. \quad (14)$$

is verified.

- ii) Deduce in particular that $E(a, b)$ admits weak existence and uniqueness in law (without explosion) in the three following cases.

- a) $b = 0$;
- b) Σ is upper bounded;
- c) $\frac{b}{a^2}$ is integrable.

- iii) Suppose a of class C^1 . Find necessary and sufficient conditions so that the Stratonovich type equation

$$X_t = X_0 + \int_0^t a(X_s) \circ dW_s$$

has a (unique) non-exploding solution (Itô process).

22. We consider the equation

$$X_t = 3 \int_0^t X_s^{\frac{1}{3}} ds + 3 \int_0^t X_s^{\frac{2}{3}} dW_s. \quad (15)$$

- i) Verify if the conditions of Theorem 13.29 (Yamada-Watanabe) concerning pathwise uniqueness are verified.
- ii) Show that (15) admits an infinity of solutions of the type

$$X_t^{(\theta)} = \begin{cases} 0 & : 0 \leq t < \beta_\theta; \\ W_t^3 & : \beta_\theta \leq t < \infty, \end{cases}$$

where $0 \leq \theta < \infty$ and

$$\beta_\theta = \inf\{s \geq \theta | W_s = 0\}.$$

- 23. Let S be a square Bessel process $\text{BESQ}(x_0, \delta)$ with $\delta > 0$. Supposing that S is strictly positive determine the SDE fulfilled by $X = \sqrt{S}$.
- 24. Let (S_t) be a $\text{BESQ}(x_0, \delta)$. For $N > 0$, we set $\tau_N = \inf\{s \geq 0 | S_s \geq N\}$.
 - (a) Show that a.s. (τ_N) is an increasing sequence of stopping times diverging to infinity.
 - (b) Show that, whenever N is large enough $E(S_t^{\tau_N}) = x_0 + \delta E(\tau_N \wedge t)$.
 - (c) Calculate $E(S_t), t \geq 0$.
 - (d) Show that $E(\sup_{t \leq T} S_t^2) < \infty$.
 - (e) Calculate $\text{Var}(S_t)$ for any $t \geq 0$.
- 25. Deduce $E(S_t)$ if S is a Square Bessel process using Theorem 14.12.

26. Let $S^i, i = 1, 2$ be the solution (on some filtered probability space, equipped with a Brownian motion W) of

$$S_t^i = x_i + 2 \int_0^t \sqrt{S_s^i} dW_s + \delta_i t, i = 1, 2,$$

where $0 \leq x_1 \leq x_2, 0 \leq \delta_1 \leq \delta_2$.

- (a) Show that $S_t^1 \leq S_t^2, \forall t \in [0, T]$.
- (b) Express $E(|S_t^1 - S_t^2|)$.
- (c) Show that

$$\text{Var}(S_t^2 - S_t^1) \leq 4(x_2 - x_1)t + 2(\delta_2 - \delta_1)t$$

- (d) Using Doob's inequality find an upper bound for

$$E \left(\sup_{t \leq T} |S_t^2 - S_t^1|^2 \right).$$

- (e) Let (x_n) (resp. δ_n) be an increasing sequence of non-negative numbers converging to x_0 (resp. δ_0). Let $S = S^n$ the solution to

$$S_t = x_n + 2 \int_0^t \sqrt{S_s} dW_s + \delta_n t, \quad n \in \mathbb{N}.$$

Deduce that the sequence S^n converges ucp to S^0 .

- (f) Let us adopt the notations of Chapter 14. Deduce that $\varphi(x_n, \delta_n) \rightarrow \varphi(x, \delta)$ in the case when μ has its support on a compact interval $[0, T]$.
27. Let X be a Cox-Ingersoll-Ross process. Determine the Laplace transform of the law of X_t i.e. $E(e^{-\lambda X_t})$, for $\lambda > 0$.